

Financial Values, Fundamental Ratios & Classical Machine Learning

Determining Stock Movement 1, 2, and 3 Days After Earnings Calls

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Project Background



Quarterly Filings

Public companies file quarterly earnings reports detailing essential financials.

- Revenue & Net Income
- Expenses & Shares
- SEC Compliance



Market Influence

Stock trading behavior is heavily influenced by these financial values and their derived ratios.

- P/E, Debt/Equity, Assets / Equity
- Changes in expenses / net income / assets, liabilities etc



The Objective

Predict stock movement 1, 2, and 3 days post-announcement using classical machine learning.

- Harness classical models
- Predict Stock Movement
- Understand how to gain competitive edge while minimizing risk

The Data



Scope & Scale

Extensive historical dataset covering various market capitalizations.

- 21,352 Datapoints (5 years)
- 1,300+ Companies
- Large, Mid, & Small Cap
- Sourced from [ROC.ai](#) API



Core Metrics

Key financial indicators extracted from quarterly earnings reports.

- Profit, Income, EBITDA
- Assets, Liabilities, Equity
- Quarterly Delta Changes



Feature Engineering

Derived financial ratios used to provide predictive power for ML models.

- EPS & P/E Ratios
- Debt-to-Equity / Assets
- Return on Equity (ROE)

Methodology & Tuning



Data Preprocessing

Z-score scaling was applied to highlight deviations from average, which are most likely to influence stock price increases.



Model Selection

Tested SVM, Logistic Regression, Adaboost, and K-NN. 12 final models overall, each trained for 1, 2, or 3 day movement predictions.



Optimization Metric

Precision for class "1" was used to minimize false positives of stock increases, effectively reducing financial risk.



Target Variables (Positive Class is Increase, Negative Class is Decrease)

- 1 Day Price Change
- 2 Day Price Change
- 3 Day Price Change

Hypothesis



Top Performing Model: SVM

We believe that the SVM will perform the best across the board. This expectation is rooted in the mathematical foundation of the algorithm:

- **Margin Maximization:** SVM tries to maximize the margin between the two classes.
- **Risk Reduction:** This approach can lead to a decrease in false positives.
- **High Fidelity:** It effectively increases precision in predictions.

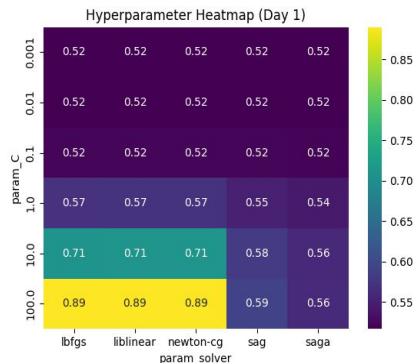
Logistic Regression: Tuning & Final Params

1

1-Day Movement

C: 100

Algo: lbfgs

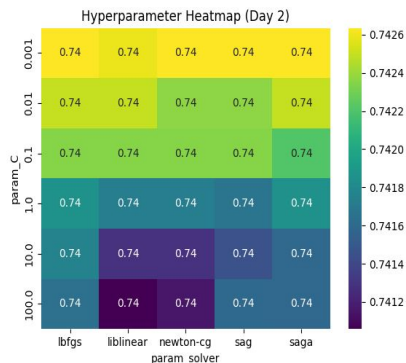


2

2-Day Movement

C: 100

Algo: lbfgs

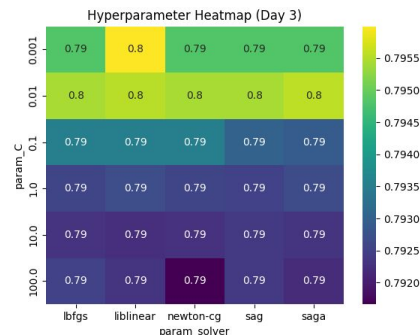


3

3-Day Movement

C: 0.001

Algo: liblinear



Logistic Regression Results

1

1-Day Prediction

Validation Accuracy: 93.05%

Test Accuracy: 94.50%

Precision: 0.90

2

2-Day Prediction

Validation Accuracy: 73.46%

Test Accuracy: 72.72%

Precision: 0.75

3

3-Day Prediction

Validation Accuracy: 78.48%

Test Accuracy: 78.17%

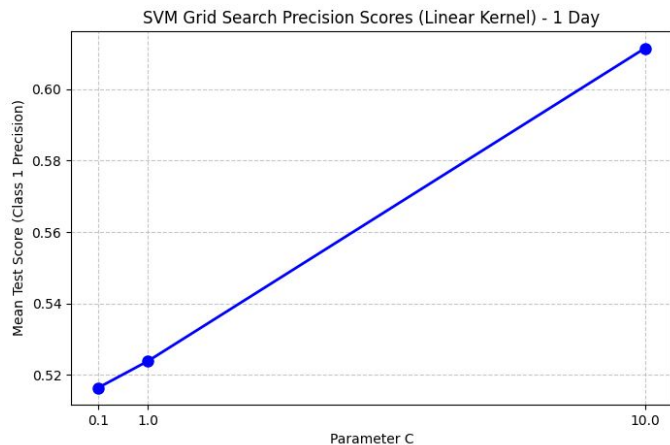
Precision: 0.79

SVM: Tuning & Final Params

Best Performing Kernel: Linear

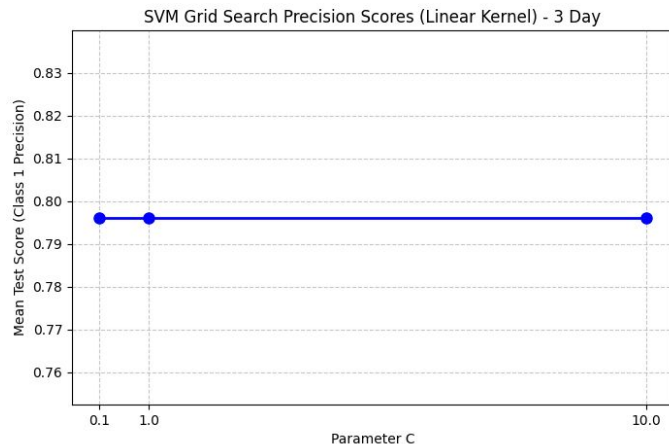
1 & 2 Day Movement

Optimal C: 10



3 Day Movement

Optimal C: 1



SVM Results

1

1-Day Prediction

Validation Accuracy: 69.81%

Test Accuracy: 72.99%

Precision: 0.64

2

2-Day Prediction

Validation Accuracy: 73.42%

Test Accuracy: 72.72%

Precision: 0.75

3

3-Day Prediction

Validation Accuracy: 78.51%

Test Accuracy: 78.26%

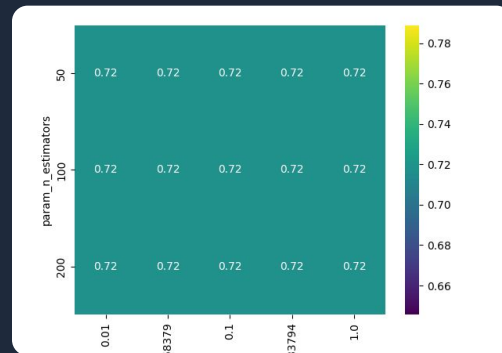
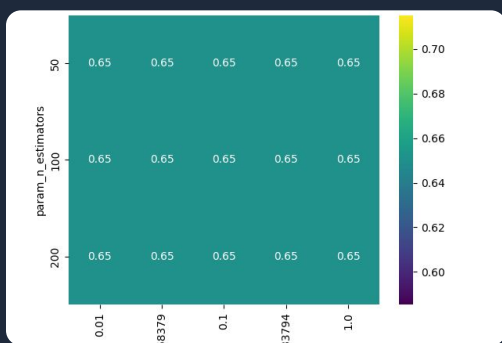
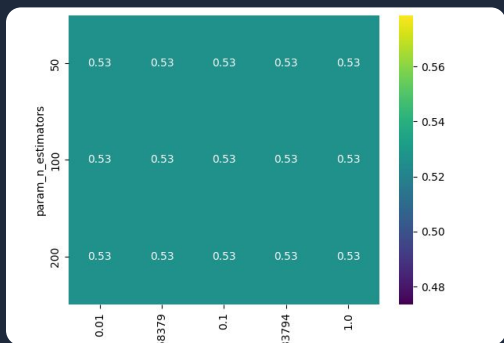
Precision: 0.79

Adaboost: Tuning & Final Params

Key Observation:

Something unique happened: every parameter set for every model produced the same result.

Decision: **Occam's Razor** — Chose the simplest model configuration.



Adaboost Results

1

1-Day Prediction

Validation Accuracy: 56.80%

Test Accuracy: 56.63%

Precision: 0.53

2

2-Day Prediction

Validation Accuracy: 62.90%

Test Accuracy: 62.37%

Precision: 0.65

3

3-Day Prediction

Validation Accuracy: 71.45%

Test Accuracy: 71.45%

Precision: 0.73

Decision Tree: Tuning & Final Params

1

1-Day Prediction

Optimal Parameters

Decision tree depth of 3 produced the most accurate results for this window.

Precision: 0.53

2

2-Day Prediction

Optimal Parameters

A shallower tree depth of 2 was sufficient to achieve peak performance.

Precision: 0.73

3

3-Day Prediction

Optimal Parameters

A much deeper tree depth of 10 was required to capture the best precision.

Precision: 0.77

Decision Tree Results

1

1-Day Prediction

Validation Accuracy: 51.97%

Test Accuracy: 50.89%

Precision: 0.53

2

2-Day Prediction

Validation Accuracy: 76.40%

Test Accuracy: 72.04%

Precision: 0.74

3

3-Day Prediction

Validation Accuracy: 76.10%

Test Accuracy: 75.88%

Precision: 0.77

Summary of Model Results

1

1-Day Movement

Top Performer: Logistic Regression

Performed miles better than any other model.

Precision: 0.90

Test Accuracy: 94.50%

2

2-Day Movement

Result: Exact Tie

SVM and Logistic Regression performed identical for this timeframe.

Precision: 0.75

Test Accuracy: 72.72%

3

3-Day Movement

Top Performer: SVM

SVM narrowly beat out Logistic Regression on this movement window.

Precision: 0.79

Test Accuracy: 78.26%

Conclusion



Logistic Regression

A strong buy signal for systematic traders targeting 1-day post-call stock price movement.



SVM Model

Effective as a secondary indicator within a basket of signals for 2 and 3-day movement strategies.



Trader's Toolbelt

While not for standalone buy decisions, these models serve as powerful analytical tools for traders.

The Journey



Initial Semantic Focus

Aimed to categorize earnings calls based on pure semantics into three distinct categories.



LongTransformer Performance

Research revealed that LongTransformer models performed only slightly above random selection.



Return to Classical Models

Pivoted back to classical models. Explored 6 categories but faced data point limitations.



Future Research Directions

Potential in combining classical models with fine-tuned LongTransformers for financial calls.